

Summary of Compound data acquisition (ChEMBL) Jupyter Notebook

The aim of this notebook is to learn about some theoretical bioactivity measures like IC50, pIC50.

In practical side, i had used some python libraries used in Cheminformatics, like *chembl_webresource_client* and *rdkit.Chem*, in addition to Data science python's library like *pandas* and *numpy* for manipulation and cleaning Data collected from ChEMBL Database.

Summary of practical work

1. Connect to ChEMBL database (creating ChEMBL_resources object)
2. Get Target data (protein data)
 - a. Query data from ChEMBL
 - b. Download data from ChEMBL
 - c. Select our target (target ChEMBL ID)
3. Get bioactivity Data
 - a. Query data from ChEMBL
 - b. Download data from ChEMBL
 - c. Preprocessing and filter bioactivity data
4. Get compound Data
 - a. Query data from ChEMBL
 - b. Download data from ChEMBL
 - c. Preprocessing and filter bioactivity data (add canonical smiles)
5. Output (bioactivity-compound) data
 - a. Merge bioactivities and compounds dataframes
 - b. Add pIC50 values
 - c. Plot pIC50 data histogram
 - d. Transform canonical smiles to rdkit object
6. Save file into a csv